Apply the Double-Angle Formulas

Formulas for $\sin 2\theta$, $\cos 2\theta$, and $\tan 2\theta$ may be obtained by applying sum and difference formulas.

$$\sin 2\theta = \sin(\theta+\theta) = \sin\theta\cos\theta + \cos\theta\sin\theta = 2\sin\theta\cos\theta \qquad \sin(u+v) = \sin u\cos v + \cos u\sin v$$

$$\cos 2\theta = \cos(\theta+\theta) = \cos\theta\cos\theta - \sin\theta\sin\theta = \cos^2\theta - \sin^2\theta \qquad \cos(u+v) = \cos u\cos v - \sin u\sin v$$

$$\tan 2\theta = \tan(\theta+\theta) = \frac{\tan\theta+\tan\theta}{1-\tan\theta\tan\theta} = \frac{2\tan\theta}{1-\tan^2\theta} \qquad \tan(u+v) = \frac{\tan u+\tan v}{1-\tan u\tan v}$$

Double-Angle Formulas

$$\sin 2\theta = 2\sin\theta\cos\theta \qquad \tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

$$\cos 2\theta = \cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta = 2\cos^2\theta - 1$$

The double-angle formulas can also be used with angles other than 2θ.

For example: $\cos 4\theta = 1 - 2\sin^2 2\theta$ $\qquad \sin A = 2\sin\frac{A}{2}\cos\frac{A}{2}$

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1. Given $\sin\theta = \frac{5}{13}$ and $\tan\theta < 0$, find $\sin 2\theta$.

2. Given $\tan\theta = -\frac{5}{4}$ and θ in Q IV, find $\cos 2\theta$.

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Verify the identity.

3. $\dfrac{\cos 2x}{1-\sin 2x} = \dfrac{\cos x + \sin x}{\cos x - \sin x}$

4. $\dfrac{\sin 3x}{\sin x} = 4\cos^2 x - 1$

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Apply the Power-Reducing Formulas

The double-angle formulas for cosine can be used to express even powers of sine and cosine in terms of the first power of cosine only.

Power-Reducing Formulas: $\sin^2\theta = \dfrac{1-\cos 2\theta}{2}$ $\cos^2\theta = \dfrac{1+\cos 2\theta}{2}$ $\tan^2\theta = \dfrac{1-\cos 2\theta}{1+\cos 2\theta}$

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5. Write $\sin^4 2x$ in terms of first powers of cosine.

Apply the Half-Angle Formulas

Half-angle formulas can be derived from the power-reducing formulas.

Substituting $\dfrac{\alpha}{2}$ for θ in the power-reducing formulas results in the half-angle formulas.

Half-Angle Formulas

$$\sin\frac{\alpha}{2} = \pm\sqrt{\frac{1-\cos\alpha}{2}} \qquad \cos\frac{\alpha}{2} = \pm\sqrt{\frac{1+\cos\alpha}{2}}$$

$$\tan\frac{\alpha}{2} = \pm\sqrt{\frac{1-\cos\alpha}{1+\cos\alpha}} = \frac{1-\cos\alpha}{\sin\alpha} = \frac{\sin\alpha}{1+\cos\alpha}$$

The sign + or − is determined by the quadrant of angle $\dfrac{\alpha}{2}$.

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6. Use the half-angle formula to find the exact value of $\sin\dfrac{5\pi}{12}$.

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7. If $\sin\alpha = -\frac{7}{25}, \frac{3\pi}{2} < \alpha < 2\pi,$ then find the exact value of $\tan\frac{\alpha}{2}.$

8. Verify the identity. $\cos^2\frac{\alpha}{2}\tan\frac{\alpha}{2} = \frac{1}{2}\sin\alpha$

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